Formula Notes - ISR Integral Basis Extension

Wenhao Ya

1 Motivation

The radiator function is

$$F(x,W) = \beta x^{\beta-1}(1+\delta) - \beta(1-\frac{x}{2}) + \frac{1}{8}\beta^2\left[4(2-x)\log\frac{1}{x} - \frac{1+3(1-x)^2}{x}\log(1-x) - 6 + x\right] \tag{1}$$

The ISR-corrected cross section can be calculated as

$$\sigma^{\mathrm{ISR}} = \int_0^b \mathrm{d}x F(x,W)\sigma(W\sqrt{1-x}), \tag{2}$$

where $b = 1 - (\frac{W_{\min}}{W})^2 \in (0,1)$, and $\sigma(W)$ is the pre-correction cross section (Born-level or dressed).

Liu et al. (2024) discussed the cases where $\sigma(W\sqrt{1-x})$ includes the Breit-Wigner propagator $\frac{1}{A+Bx}$ and the W^{2m} factors $(1-x)^m$ with integer m. All possible integrals can be written as linear superposition of the basis $\{\int \mathrm{d}x F\cdot(1+x)^m, \int \mathrm{d}x \frac{F}{A+Bx}\}$.

In this note, in order to make the Taylor expansion of a correction function $\xi(W)$ possible to be multiplied into the integrand, we generalize it a little further, including the W^{-n} factors $(1-x)^{-\frac{n}{2}}$ (the negative sign is just a convention). It turns out that we have to add this set into the basis: $\{\int \mathrm{d}x \frac{F}{(1-x)^{\frac{n}{2}}}, \int \mathrm{d}x \frac{F\sqrt{1-x}}{A+Bx}\}$. The $\int \mathrm{d}x \frac{F}{(1-x)^{\frac{n}{2}}}$ family is easy to get, in terms of special functions and realized by series expansion. However, the $x^{\beta-1}$ and $\log(x)$ contributions to the $\int \mathrm{d}x \frac{F\sqrt{1-x}}{A+Bx}$ integral can only be expressed by the Appell's F_1 function and its derivative, which involve double series and are hard to handle. In this note we try to solve this.

2 Raw Integrals (Given by Mathematica)

$$\begin{aligned}
I_{9,\mathrm{raw}} &= \int \mathrm{d}x \frac{F}{(1-x)^{-1/2}} \\
&= \frac{1}{900}\beta\Big\{ -480 + (1-b)^{3/2}[-9b(20+3\beta) + 8(60+79\beta)] + 8\beta(-79+60\gamma+15i\pi) \\
&+ 600\beta\mathcal{B}(1-b,\frac{5}{2},0) + 900(1+\delta)\mathcal{B}(b,\beta,\frac{3}{2}) + 15\beta\Big[15\pi^2 - 8(\frac{11}{3}+4\gamma+\log 256) \\
&+ 30\mathcal{B}(1-b,\frac{3}{2},0)\log(1-b) + (1-b)^{3/2}\Big(-\frac{16(\sqrt{-1+b}+\mathrm{ArcCsc}(\sqrt{b}))}{(-1+b)^{3/2}} \\
&- 30\Phi(1-b,2,\frac{3}{2}) + (-8+3b)(3\log(1-b) - 4\log b)\Big)\Big]\Big\}
\end{aligned} \tag{3}$$

$$
\begin{aligned}
I_{7,\mathrm{raw}} &= \int \mathrm{d}x \frac{F}{(1-x)^{1/2}} \\
&= \frac{1}{36}\beta\Big\{48(-1+\sqrt{1-b}) - \sqrt{1-b}\; b(12+\beta) + \beta(-4+4\sqrt{1-b}+9\pi^2) \\
&+ 48i\beta\mathrm{ArcSec}(\sqrt{b}) + 36(1+\delta)\mathcal{B}(b,\beta,\frac{1}{2}) - 18\sqrt{1-b}\;\beta\Phi(1-b,2,\frac{1}{2}) + 3\beta\Big[-16\log 4 \\
&+ 12\mathrm{ArcTanh}(\sqrt{1-b})(4+\log(1-b)) + \sqrt{1-b}(-4+b)(3\log(1-b)-4\log b)\Big]\Big\}
\end{aligned} \tag{4}
$$

$$
\begin{aligned}
I_{8,\mathrm{raw}} &= \int \mathrm{d}x \frac{F}{(1-x)^{3/2}} \\
&= \frac{1}{4}\beta\Big\{\frac{b(-4+\beta)}{\sqrt{1-b}} + \beta(8+\pi^2) - 16i\beta\mathrm{ArcSec}(\sqrt{b}) - 16\beta\mathrm{ArcTanh}(\sqrt{1-b}) \\
&+ 4(1+\delta)\mathcal{B}(b,\beta,-\frac{1}{2}) - \frac{\beta}{\sqrt{1-b}}\Big[2\Phi(1-b,2,-\frac{1}{2}) + \Big(-3b \\
&+ 4\;{}_2F_1(-\frac{1}{2},1,\frac{1}{2},1-b)\Big)\log(1-b) + 4b\log b\Big]\Big\}
\end{aligned} \tag{5}
$$

$$
\begin{aligned}
I_{5,\mathrm{raw}} &= \int \mathrm{d}x \frac{F}{(1-x)^{5/2}} \\
&= \frac{1}{36}\beta\Big\{48 + 84\beta + \frac{-48-52\beta+9b(4+3\beta)}{(1-b)^{3/2}} - 24\beta\gamma + 24i\beta\pi + 9\beta\pi^2 \\
&- 24\beta\mathcal{B}(1-b,-\frac{1}{2},0) + 36(1+\delta)\mathcal{B}(b,\beta,-\frac{3}{2}) + 24\beta\log 4 \\
&+ 18\beta\mathcal{B}(1-b,-\frac{3}{2},0)\log(1-b) - \frac{i\beta}{(-1+b)^{3/2}}\Big[48(-1+b)^{3/2}\mathrm{ArcCsc}(\sqrt{b}) \\
&- 18\Phi(1-b,2,-\frac{3}{2}) - (-4+3b)(16+9\log(1-b)-12\log b)\Big] - 24\beta\psi(\frac{5}{2})\Big\}
\end{aligned} \tag{6}
$$

$$
\begin{aligned}
I_{10,\mathrm{raw}} &= \int \mathrm{d}x \frac{F}{(1-x)^4} \\
&= \frac{1}{288(1-b)^3}\beta\Big\{-b\Big[288 + 120(-3+b)b + 222\beta + b(-249+91b)\beta\Big] \\
&+ 48b^{\beta}(1+\delta)\Big[11 + b^2(-3+\beta)(-2+\beta) + (-6+\beta)\beta - b(-3+\beta)(-5+2\beta)\Big] \\
&+ 48(-1+b)^3(-3+\beta)(-2+\beta)(-1+\beta)(1+\delta)\mathcal{B}(b,\beta,0) \\
&+ 6\beta\Big[40b^3\mathrm{ArcTanh}(1-2b) + \log(1-b)\Big(-49 + 3(37-28b)b - 12(-1+b)^3\log(1-b)\Big) \\
&+ 12b(-4+5b)\log(b)\Big] - 144(-1+b)^3\beta\;\mathrm{Li}_2(b)\Big\}
\end{aligned} \tag{7}
$$

$$
\begin{aligned}
I_{11,\text{raw}} &= \int \mathrm{d}x \frac{F}{(1-x)^{7/2}} \\
&= \frac{1}{900}\beta\Bigg\{480 + \frac{25b(12+5\beta) - 8(60+43\beta)}{(1-b)^{5/2}} + 8\beta(43 - 60\gamma + 15i\pi) \\
&- 360\beta\ \mathcal{B}(1-b, -\frac{3}{2}, 0) + 900(1+\delta)\mathcal{B}(b, \beta, -\frac{5}{2}) + \beta\Bigg[2072 + 225\pi^2 \\
&+ 450\mathcal{B}(1-b, -\frac{5}{2}, 0)\log(1-b) - \frac{i}{(-1+b)^{5/2}}\Bigg(240(-1+b)^{5/2}\mathrm{ArcCsc}(\sqrt{b}) \\
&+ 450\Phi(1-b, 2, -\frac{5}{2}) + 45(-8+5b)\log(1-b) \\
&- 4(92 + 20b(-7+3b) + 15(-8+5b)\log(b))\Bigg) - 360\psi(-\frac{3}{2}) - 120\psi(\frac{7}{2})\Bigg]\Bigg\} \qquad (8)
\end{aligned}
$$

$\mathcal{B}(z, a, b)$ is the incomplete beta function, ${}_2F_1$ is the hypergeometric function, $\Phi(z, a, b)$ is the Lerch transcendent, and $\psi(z)$ is the digamma function. γ is Euler's gamma constant.

Verifications by Mathematica are shown in Fig. 1.

3 Simplified Integrals

For better numerical stability, all the integrals are simplified as follows.

$$
\begin{aligned}
I_9 &= \int \mathrm{d}x \frac{F}{(1-x)^{-1/2}} \\
&= \frac{\beta(1+\delta)}{1+2\beta}\left[2\sqrt{1-b}\ b^{\beta} + \mathcal{B}(b, \beta, \frac{1}{2})\right] + \frac{1}{900}\beta\Bigg\{-480 \\
&+ (1-b)^{3/2}\Big[480 - 180b + (232-27b)\beta\Big] + \beta\Bigg[-1072 + 840\sqrt{1-b} + 225\pi^2 \\
&+ 960\mathrm{ArcTanh}(\sqrt{1-b}) - 480\log 4 + 900\Big(-\sqrt{1-b} + \mathrm{ArcTanh}(\sqrt{1-b})\Big)\log(1-b) \\
&+ 15(1-b)^{3/2}(-8+3b)\Big(3\log(1-b) - 4\log(b)\Big) \\
&- 900\Big(-\mathrm{Li}_2(-\sqrt{1-b}) + \mathrm{Li}_2(\sqrt{1-b})\Big)\Bigg]\Bigg\} \qquad (9)
\end{aligned}
$$

$$
\begin{aligned}
I_7 &= \int \mathrm{d}x \frac{F}{(1-x)^{1/2}} \\
&= \frac{1}{36}\beta\Bigg\{48(-1+\sqrt{1-b}) - \sqrt{1-b}\; b(12+\beta) + \beta(-4+4\sqrt{1-b}+9\pi^2) \\
&+ 36(1+\delta)\mathcal{B}(b,\beta,\frac{1}{2}) + 36\beta\mathrm{ArcTanh}(\sqrt{1-b})\log(1-b) + 3\beta\Bigg[32\mathrm{ArcTanh}(\sqrt{1-b}) \\
&- 16\log 4 + \sqrt{1-b}(-4+b)\Big(3\log(1-b) - 4\log b\Big)\Bigg] \\
&- 36\beta\Big[-\mathrm{Li}_2(-\sqrt{1-b}) + \mathrm{Li}_2(\sqrt{1-b})\Big]\Bigg\}
\end{aligned} \tag{10}
$$

$$
\begin{aligned}
I_8 &= \int \mathrm{d}x \frac{F}{(1-x)^{3/2}} \\
&= \frac{1}{4}\beta\Bigg\{-4(-1+2\beta)(1+\delta)\mathcal{B}(b,\beta,\frac{1}{2}) + \frac{1}{\sqrt{1-b}}\Bigg[b(-4+\beta) + 8(-1+\sqrt{1-b})\beta \\
&+ 8b^{\beta}(1+\delta) + \sqrt{1-b}\;\beta\pi^2 + (-4+3b)\beta\log(1-b) + 4\sqrt{1-b}\;\beta\mathrm{ArcTanh}(\sqrt{1-b})\log(1-b) \\
&- 4b\beta\log b - 4\sqrt{1-b}\;\beta\Big(-\mathrm{Li}_2(-\sqrt{1-b}) + \mathrm{Li}_2(\sqrt{1-b})\Big)\Bigg]\Bigg\}
\end{aligned} \tag{11}
$$

$$
\begin{aligned}
I_5 &= \int \mathrm{d}x \frac{F}{(1-x)^{5/2}} \\
&= \beta(1+\delta)\Bigg\{\frac{2b^{\beta}}{3(1-b)^{3/2}} + (-3+2\beta)\Bigg[-\frac{2b^{\beta}}{3\sqrt{1-b}} + \frac{1}{3}(-1+2\beta)\mathcal{B}(b,\beta,\frac{1}{2})\Bigg]\Bigg\} \\
&+ \frac{1}{36}\beta\Bigg\{48 + \frac{-48-52\beta+9b(4+3\beta)}{(1-b)^{3/2}} - \frac{\beta(24+36\log(1-b))}{\sqrt{1-b}} \\
&+ \beta\frac{-8-12\log(1-b)+(4-3b)(16+9\log(1-b)-12\log b)}{(1-b)^{3/2}} \\
&+ \beta\Bigg[20+9\pi^2+48\log 4 + 36\mathrm{ArcTanh}(\sqrt{1-b})\Big(-\frac{8}{3}+\log(1-b)\Big) \\
&- 36\Big(-\mathrm{Li}_2(-\sqrt{1-b}) + \mathrm{Li}_2(\sqrt{1-b})\Big)\Bigg]\Bigg\}
\end{aligned} \tag{12}
$$

$$
\begin{aligned}
I_{10} &= \int \mathrm{d}x \frac{F}{(1-x)^4} \\
&= \beta(1+\delta)\Bigg\{\frac{b^\beta}{3(1-b)^3} + (-3+\beta)\Bigg[-\frac{b^\beta}{6(1-b)^2} + (-2+\beta)\Bigg(\frac{b^\beta}{6(1-b)} - \frac{-1+\beta}{6}\mathcal{B}(b,\beta,0)\Bigg)\Bigg]\Bigg\} \\
&+ \frac{1}{288(1-b)^3}\beta\Bigg\{-48 - 72(1-b) + 120(1-b)^3 \\
&+ \beta\Bigg[-64 - 3(1-b) - 24(1-b)^2 + 91(1-b)^3 \\
&+ 6\Bigg(\big[20(1-b)^3 - 12(1-b) - 8\big]\log b - \big[2 + 3(1-b) + 24(1-b)^2 + 20(1-b)^3\big]\log(1-b) \\
&+ 12(1-b)^3\log^2(1-b)\Bigg) + 144(1-b)^3\mathrm{Li}_2(b)\Bigg]\Bigg\} \qquad (13)
\end{aligned}
$$

$$
\begin{aligned}
I_{11} &= \int \mathrm{d}x \frac{F}{(1-x)^{7/2}} \\
&= \beta(1+\delta)\Bigg\{\frac{2b^\beta}{5(1-b)^{5/2}} + (-5+2\beta)\Bigg[-\frac{2b^\beta}{15(1-b)^{3/2}} + (-3+2\beta)\Bigg(\frac{2b^\beta}{15\sqrt{1-b}} \\
&- \frac{-1+2\beta}{15}\mathcal{B}(b,\beta,\frac{1}{2})\Bigg)\Bigg]\Bigg\} + \frac{1}{900}\beta\Bigg\{480 - \frac{300}{(1-b)^{3/2}} - \frac{180}{(1-b)^{5/2}} \\
&+ \beta\Bigg[1088 + 225\pi^2 + 480\log 4 - 960\mathrm{ArcTanh}(\sqrt{1-b}) + 900\mathrm{ArcTanh}(\sqrt{1-b})\log(1-b) \\
&- 900\Big(\mathrm{Li}_2(\sqrt{1-b}) - \mathrm{Li}_2(-\sqrt{1-b})\Big) - \frac{840 + 900\log(1-b)}{\sqrt{1-b}} \\
&- \frac{5 + 75\log(1-b) + 300\log b}{(1-b)^{3/2}} - \frac{243 + 45\log(1-b) + 180\log b}{(1-b)^{5/2}}\Bigg]\Bigg\} \qquad (14)
\end{aligned}
$$

For the derivation of I_6, please see another note *Propagator Contribution to the Extended ISR Integral Basis*. Here we just paste the results.

$$
\begin{aligned}
I_6 &= \int_0^b \mathrm{d}x \frac{F(x,W)\sqrt{1-x}}{A+Bx} \\
&= I_A + I_B \\
&= \frac{\beta}{24\ A\ B^{5/2}}[2C_A + 3\beta C_B] \\
&+ \beta(1+\delta)\Bigg[\frac{b^\beta}{A\beta}\cdot\frac{\beta\pi}{\sin(\beta\pi)}(-y)^{-\beta}\sqrt{1-\frac{b}{y}} - \frac{\beta}{y}\sum_{q=0}^{\infty}\frac{1}{y^q(\beta-1-q)}\sum_{k=0}^{\infty}\frac{(\beta-1-q)(-\frac{1}{2})_k}{(\beta-1-q+k)}\frac{b^k}{k!}\Bigg] \\
&+ \frac{\beta^2 b}{Ay}\sum_{q=0}^{\infty} w_q\Bigg[-\Big(\log b\ \cdot\ H_q(y) + S_q(y)\Big) + \frac{b}{4y}\Big((\log b + \frac{1}{2})H_{q+1}(y) + 2S_{q+1}(y)\Big)\Bigg]
\end{aligned}
\qquad (15)
$$

With the definitions

$$
u = \sqrt{1-b}, \quad \Delta = 1-u, \quad L_b = \log(1-b), \quad L_u = \log(1+u) \qquad (16)
$$

$$s = \sqrt{B}, \quad B^{3/2} = Bs, \quad B^{5/2} = B^2 s \tag{17}$$

$$D = A + B, \quad S = \sqrt{D}, \quad r_0 = \frac{s}{S}, \quad r_1 = \frac{su}{S} \tag{18}$$

$$z_0 = \frac{B}{D}, \quad z_1 = \frac{B(1-b)}{D} = z_0(1-b) \tag{19}$$

$$P = 8B + 6B\beta + A(4+\beta) = 2B(4+3\beta) + A(4+\beta), \quad Q = 3A^2 + 6AB + 4B^2 \tag{20}$$

$$\begin{aligned} C_A = {} & 3A^2 s\Delta(4+7\beta) + ABs\left[ub(4+3\beta) + \Delta(28+57\beta)\right] + B^{5/2}\beta\pi^2 \\ & - 3\Big[A\ S\ P\ \mathrm{ArcTanh}(r_0) + \beta L_b\left(A\ s\ u((b-7)B - 3A) - 2B^{5/2}L_u\right) \\ & + S\ \mathrm{ArcTanh}(r_1)(-A\ P + Q\ \beta\ L_b)\Big] \end{aligned} \tag{21}$$

$$C_B = 8B^{5/2}\left[\mathrm{Li}_2(1-u) + \mathrm{Li}_2(-u)\right] + S\ Q[\mathrm{Li}_2(z_0) - 4\mathrm{Li}_2(r_0) + 4\mathrm{Li}_2(r_1) - \mathrm{Li}_2(z_1)] \tag{22}$$

$$H_m = -\log(1-y) - \sum_{k=1}^{m} \frac{y^k}{k}, \quad S_m = \sum_{k=1}^{m} \frac{y^k}{k^2} - \mathrm{Li}_2(y), \quad w_q = \frac{(-\frac{1}{2})_q}{q!}\left(\frac{b}{y}\right)^q \tag{23}$$

4 C++ Implementation Performance

The benchmark test results are listed in Fig. 2. In the benchmark W is scanned over the range $[3.04\ \mathrm{GeV}, 3.13\ \mathrm{GeV}]$ to get the averaged running time of the evaluation of each integral. The $I_5,\ I_7,\ I_8,\ I_9$ are all computed in one call and the averaged time for evaluation is $\sim 0.37\ \mu$s. The computation of I_6 involves double series expansion of Appell's F_1 function so can be much slower, averaged $\sim\ 5\ \mu$s.

```
(* F/(1-x)^(-1/2) *)
NIntegrate[RadF/(1 - x)^(-1/2) /. {beta → 0.1, delta → 1}, {x, 0, 1/2}]
N[IG1Sim1 /. {beta → 1/10, delta → 1, b → 1/2}]
N[I9mod1Stable /. {beta → 1/10, delta → 1, b → 1/2}]

Out[149]= 1.78787
Out[150]= 1.78787 + 7.89492 × 10^-19 i
Out[151]= 1.78787
```

```
In[152]:= (* F/(1-x)^(1/2) *)
NIntegrate[RadF/(1 - x)^(1/2) /. {beta → 0.1, delta → 1}, {x, 0, 1/2}]
N[IGn1 /. {beta → 1/10, delta → 1, b → 1/2}]
N[I7mod1Stable /. {beta → 1/10, delta → 1, b → 1/2}]

Out[152]= 1.87617
Out[153]= 1.87617 + 0. i
Out[154]= 1.87617
```

```
In[155]:= (* F/(1-x)^(3/2) *)
NIntegrate[RadF/(1 - x)^(3/2) /. {beta → 1/10, delta → 1}, {x, 0, 1/2}]
N[IGn3Sim1 /. {beta → 1/10, delta → 1, b → 1/2}]
N[I8mod1Stable /. {beta → 1/10, delta → 1, b → 1/2}]

Out[155]= 2.00137
Out[156]= 2.00137 + 0. i
Out[157]= 2.00137
```

```
In[158]:= (* F/(1-x)^(5/2) *)
NIntegrate[RadF/(1 - x)^(5/2) /. {beta → 1/10, delta → 1}, {x, 0, 1/2}]
N[IGn5Sim1 /. {beta → 1/10, delta → 1, b → 1/2}]
N[I5mod1Stable /. {beta → 1/10, delta → 1, b → 1/2}]

Out[158]= 2.18588
Out[159]= 2.18588 - 7.40149 × 10^-18 i
Out[160]= 2.18588
```

```
In[384]:= NIntegrate[RadF Sqrt[1 - x]/(A + B x) /. {beta → 1/20, delta → 0.1, A → AvSampleP, B → BvSampleP}, {x, 0, bbSampleP}]
Out[384]= 0.211196 + 80.5156 i
```

Figure 1: Mathematica Integral Verifications. (For the analytic value of I_6 please see Fig. 2)

```
---- Timing (totals over repeats*nW evaluations) ----
I5() only                Real= 0.0452609 s  CPU=     0.05 s  us/pt=  0.452609  checksum=93016.007869657042
I7() only                Real=  0.046459 s  CPU=     0.04 s  us/pt=   0.46459  checksum=91307.352939128104
I8() only                Real=  0.039386 s  CPU=     0.04 s  us/pt=   0.39386  checksum=92118.897226674657
I9() only                Real= 0.0403941 s  CPU=     0.04 s  us/pt=  0.403941  checksum=90570.844539814294

Separate all 4           Real=  0.181064 s  CPU=     0.18 s  us/pt=   1.81064  checksum=367013.10257526231
compute_all (4-in-1)     Real= 0.0365932 s  CPU=     0.04 s  us/pt=  0.365932  checksum=367013.10257526231

Note: 'Separate all 4' calls the single-integral wrappers four times per point.
      'compute_all' shares special-function cache, so it should be much faster.

Sanity check (b=0.5, beta=0.10000000000000001, delta=1):
I5 = 2.1858804346419092
I7 = 1.8761708701562538
I8 = 2.0013662310477325
I9 = 1.7878714117737509
```

```
--- Timing (totals over repeats*nW evaluations) ----
compute_I6 (full)        Real= 0.0999131 s  CPU=      0.1 s  us/pt=   4.99566  checksum=(979206.438325,242899.662319)  avg_q=20.7  avg_k=17.2  fails=0
compute_IA only          Real= 0.0220611 s  CPU=     0.02 s  us/pt=   1.10306  checksum=(-1145.8898244, 2693.7155250i)
IromI only               Real= 0.0364368 s  CPU=     0.04 s  us/pt=   1.82184  checksum=(17819581.2836,4457389.37774i)  avg_q=14.2  avg_k=17.2  fails=0
I23 only                 Real= 0.0404191 s  CPU=     0.05 s  us/pt=   2.32096  checksum=(275.36953319,436.96280943i)  avg_q=14.5  avg_k=0  fails=0

Notes:
 - avg_q for compute_I6 is (Q_romI + Q_I23); avg_k is k_avg from romI inner series.
 - If |y| is not >~1, the chosen analytic form may converge slower.

Sanity check: I6 at W=3.08 GeV is approximately
(0.211,80.5)
```

Figure 2: The performance and validity test of C++ implementations of the integrals